

# The joint potential, nonorthogonal vacua and the protection test

263 allowed scalar contractions, physical weak CP and four independent observable directions

PerfectPower research draft / 6 October 2026 (UTC)

Fully minimized nonorthogonal vacua escape the orthogonal-frame bound. Independent couplings still move all four mixing observables. Observed CKM values and the golden relation are not predicted.

The nonorthogonal route is dynamically possible in the declared field content. This calculation enumerates the entire independent CP-even scalar polynomial space through degree six, activates every coefficient, and solves the full 54-real-field stationarity equations. It obtains positive local spectra, small nonzero mixing and physical weak CP. A shared-heavy-ray example also gives the ordering  $|V_{ub}| < |V_{cb}| < |V_{us}|$ . Neither example reproduces the measured CKM matrix, derives the quark-column hierarchy, selects a global vacuum or predicts a golden coefficient.

The decisive constraint comes from the allowed parameter space. On the retained nonorthogonal branch, independent scalar and quark couplings generate four independent physical observable directions, including after compensating the weights to hold all six quark masses fixed. Thus no exact nontrivial relation among the three mixing magnitudes and  $J$  holds throughout a neighborhood of this branch with these couplings free. Fully re-minimized one-percent scalar deformations directly change the effective coefficient, also after mass matching. The remaining task is to derive correlations among the interactions and quark weights, rather than extend an unconstrained potential search.

## Field content, order and conventions

Keep the preceding  $G_u \times G_d$  construction, with each factor the genuine  $3.A_6$  triplet group. The physical row link  $L$  transforms as  $g_u L g_d^\dagger$ . Three physical source vectors  $u_i$  transform under  $g_u$  and three  $d_i$  under  $g_d$ . Their chiral counterparts are conjugates. Each source has a distinct  $C_6$  shaping charge. CP acts through the previously checked involution  $X$ :  $L$  maps to  $X \text{ conjugate}(L) X^\dagger$  and each vector maps to  $X \text{ conjugate}(\phi)$ . The quark chain includes the independent holomorphic reverse cofactor vertex. Its tree matching and right-handed canonical normalization are retained.

The word complete here has a specified scope: every real CP-even polynomial scalar contraction through total field degree six, on the zero-quark, zero-Higgs flavor slice, under these two finite family groups and all six independent shaping factors. Constants are discarded. The  $F$  potentials induced by the earlier superpotential are kept as a separately specified positive completion: link degrees four, seven and ten, and source degree ten. This is a scalar symmetry-breaking EFT. General Kähler terms, all higher-degree quark operators, loop matching, electroweak breaking and a gauge/anomaly completion are not included in this census. There are infinitely many higher-order allowed terms; no claim of an all-order scalar action is made.

Use  $\rho=1$  and express every link or source field in units of  $\rho$ , taking the reference mass scale  $\Lambda=1$ . Restoring it, the fields are  $\Lambda \rho z$  and  $V_{\text{physical}} = \Lambda^4 \rho^{10} V_{\text{scaled}}$ . A degree- $d$  coefficient  $c$  in the receipt means  $c \Lambda^{(4-d)} \rho^{(10-d)}$ . The raw real coordinates are  $\text{Re}(z)$ ,  $\text{Im}(z)$ ; canonical real squared masses are  $\Lambda^2 \rho^{8/2}$  times their scaled Hessian eigenvalues. This factor is necessary because a canonical complex scalar is  $(q+ip)/\sqrt{2}$ .

## The exact joint operator census

$$N_{\text{CP even}} = 7 + 1 + 41 + 7 + 207 = 263$$

The independent counts are as follows. The complex column counts polynomials before Hermiticity and CP; CP-even plus CP-odd real dimensions equal that complex dimension.

| Field degree | CP-even real terms | Complex dimension |
|--------------|--------------------|-------------------|
| 2            | 7                  | 7                 |
| 3            | 1                  | 2                 |
| 4            | 41                 | 41                |
| 5            | 7                  | 14                |
| 6            | 207                | 223               |
| Total        | 263                | 287               |

An independent exact character calculation obtains the complex dimensions. For link bidegree  $(p,q)$ , Cauchy's decomposition expands  $\text{Sym}^p(3 \otimes \bar{3})$  and its conjugate into Schur representations. Local Jacobi-Trudi characters are calculated in  $\mathbb{Q}(\sqrt{5}, \omega)$ , and averaged exactly on each 1080-element group, with the balanced source characters multiplied in. Separate C6 charges require each source to occur equally often with its conjugate at these orders, except for an individual holomorphic sixth power. The six source sextics and their conjugates are added explicitly. The receipt retains every contributing multidegree sector, not just totals.

Independence is also certified without floating rank tests. The balanced contractions have integer coefficients in real coordinates. Evaluate them at 260 reproducible integer-coordinate points modulo the prime 1000003, using formal Gaussian residues and conjugation. Full-rank square evaluation minors establish exact independence of the 7 quadratic, 41 quartic and 199 balanced sextic polynomials. Their pivot-row witnesses are retained. Distinct source/link phase bidegrees separate the remaining source and determinant sectors. The two holomorphic link sextics are independent already on  $L = e_1 e_1^{\text{transpose}}$ , where  $I_6 = 1$  but  $\det(L)^2 = 0$ . Multiplication by the nonzero determinant polynomial preserves independence of the quadratic factors used at degree five. Together with the character dimensions and CP graph pairing, this establishes the complete independent real basis.

The explicit contraction basis is constructed independently from trace and Gram graphs. Local finite-group tensor counts agree with  $SU(3)$  through degree six on all balanced index sectors. The only additional finite-group contributions at this order are the individual source sextics and the link sextic. The group-specific sextic is therefore not used to remove any otherwise permitted balanced cross-sector interaction. CP pairs conjugate graph orientations. The two within-sector Gram triangles and the six link-dressed same-sector triangles give eight CP-odd balanced sextics; the holomorphic source and link pairs give eight more. Cubic and quintic conjugate pairs give respectively one and seven CP-odd terms. There are no CP-odd quadratic or quartic terms.

For an explicit basis, set  $M_u = L L^{\dagger}$ ,  $M_d = L^{\dagger} L$ ,  $S = \text{Tr } M_u$ ,  $T = \text{Tr } M_u^2$ ,  $R = \text{Tr } M_u^3$ ,  $D = \det L$ ,  $N_i = \phi_i^{\dagger} \phi_i$ ,  $Q_i = \phi_i^{\dagger} M_{\text{sector}} \phi_i$  and  $Q2_i = \phi_i^{\dagger} M_{\text{sector}}^2 \phi_i$ . Define  $G_{ij} = \phi_i^{\dagger} \phi_j$  only for sources in the same sector. The operator receipt provides a name, degree and coefficient for every contraction.

At degree two the basis is  $S$  and the six  $N_i$ . Degree three contains  $\text{Re } D$ . Degree four contains  $S^2$  and  $T$ , twelve  $S N_i$  and  $Q_i$  terms, all twenty-one norm products  $N_i N_j$ , and the six within-sector  $|G_{ij}|^2$ . Degree five contains  $S \text{Re } D$  and six  $N_i \text{Re } D$  terms.

At degree six the link-only terms are  $S^3$ ,  $S T$ ,  $R$ ,  $\text{Re } D^2$  and  $\text{Re } I_6$ . The twenty-four link-four/source-two terms are  $S^2 N_i$ ,  $T N_i$ ,  $S Q_i$  and  $Q2_i$ . The seventy-eight link-two/source-four terms comprise twenty-one  $S N_i N_j$ , thirty-six  $Q_i N_j$ , six  $S |G_{ij}|^2$ , six

real dressed Gram triangles and nine  $X_{ij}=|u_i\dagger L d_j|^2$ . The one hundred source-only terms are fifty-six triple norm products, thirty-six norm times within-sector overlap squares, two real Gram triangles and six  $\text{Re } F6(\phi_i)$ . A redundant  $|\det L|^2$  is eliminated by Newton's trace identity. This gives  $5+24+78+100=207$  sextics.

All nine  $X_{ij}$  have independent coefficients. Their absolute squares are neutral under any additional Abelian phase assignments to the displayed fields: the charge of  $u_i\dagger L d_j$  cancels that of its conjugate. Ordinary phase charges alone therefore cannot forbid these norm-square operators, even if they constrain holomorphic vertices. A mechanism that relates or removes them needs an additional structural restriction, such as a justified interaction origin or a symmetry acting nontrivially on source labels. No such restriction is assumed here.

## A bounded potential with every coefficient active

The scaled base potential is the squared gradient of  $W_{\text{link}}=-32.4 \det L + I6(L) + .2(\det L)^2$ , plus the sum of six source potentials  $\text{norm}(\text{grad } F6(\phi_i))^2 - 36 \text{norm}(\phi_i)^2 - 48 \text{Re } F6(\phi_i)$ . Its nonzero link  $L=I$  and all group-rotated coordinate sources reproduce the previously derived stationary branch. Sources need not be orthogonal to each other: each field independently occupies a stable axis orbit.

Add a real coefficient times every one of the 263 basis operators. The declared deterministic coefficient prescription uses seed 108066: small positive, distinct generic coefficients at the other terms, a positive .02 coefficient for  $S^3$ , independent positive within-sector overlaps near .5 and independent negative  $X_{ij}$  coefficients near -.5. These are chosen EFT inputs. Their ratios are not derived or adjusted to the golden relation. Every coefficient is nonzero in the three mixed cases; the unmixed case is retained as the baseline. A common factor 0, .25, 1 or 4 scales the added polynomial.

The positive pure-link sextic is a deliberate boundedness condition, not a flavor target. By Cauchy-Schwarz,  $|I6(L)| \leq 16 S^3$ , and  $|\text{Re}(\det L)^2| \leq S^3/27$ . Also  $S$ ,  $T$  and  $R$  are bounded by  $S^3$ . The retained pure-link sextic has a strictly positive lower-bound coefficient .0194628392 after subtracting all these absolute contributions. The established smoothness of  $F6$  implies a positive minimum of  $\text{norm}(\text{grad } F6)^2$  on the unit sphere, so it gives degree-ten radial growth for each source. Every mixed contraction has weighted growth  $a/6+b/10 < 1$ , where  $a$  and  $b$  count link and source fields. Young's inequality absorbs such terms into the positive link-six and source-ten bounds. All positive-factor potentials are consequently coercive and possess global minima. The calculations identify local minima; they do not classify those global minima.

## Nonorthogonal local vacua and physical CP

$$\text{Im}(G_{12}G_{23}G_{31}) = \frac{\sqrt{3}}{16}, \quad \det G = \frac{1}{8}$$

Start with the first full-rank axis-ray triple having a nonzero imaginary Gram triangle, and use it in both sectors. In the fixed enumeration its ray indices are 0,3,10. Exact field arithmetic gives  $G_{12} G_{23} G_{31} = (1+i \sqrt{3})/16$  and  $\det G = 1/8$ . The individual source directions are genuine group-rotated stationary axes, not a copied CKM chart. Each is stable before adding the mixed operators. The phase invariant exists despite real potential and quark coefficients.

There is a useful exact CP identity for any shared full-rank column matrix  $C$ . Write  $H_u = C \text{diag}(a_i) C^\dagger$  and  $H_d = C \text{diag}(b_i) C^\dagger$ , with real weights, and  $G = C^\dagger C$ . Direct multiplication gives  $[H_u, H_d] = C A C^\dagger$ , where  $A_{ij} = (a_i b_j - b_i a_j) G_{ij}$  and  $A_{ii} = 0$ . Therefore

$$\det[H_u, H_d] = 2i \det(G) (a_1 b_2 - b_1 a_2)(a_2 b_3 - b_2 a_3)(a_3 b_1 - b_3 a_1) \operatorname{Im}(G_{12} G_{23} G_{31}).$$

For the retained triple this is  $i \sqrt{3}/64$  times the three weight differences. The raw weighted matrices violate weak CP when that product is nonzero. On a scalar group-aligned link, the canonical up and down matching maps are strictly increasing spectral functions of these raw matrices, so their left projectors and physical  $J$  are unchanged. This proves the mechanism's CP content without identifying a complex entry with a physical phase. The perturbed link is handled by the full holomorphic mediator block and canonical matching, and CP is checked again there.

The quark-column inputs are  $(.007, .05, .9)$  in the up sector and  $(.02, .2, .85)$  in the down sector. These explicitly imposed hierarchies are not outputs of the scalar symmetry. The full 54-coordinate minimization, followed by Newton refinement, gives stationarity residuals below  $4e-11$  across the four cases. Their raw Hessian minima range from 108 at the baseline to 102.70 at factor four. The independent weights preserve three nondegenerate quark modes in each sector.

At factor one the matched magnitudes are  $|V_{us}|=.00139934$ ,  $|V_{cb}|=.01621072$  and  $|V_{ub}|=.00038814$ , with  $J=8.64911e-9$  and standard phase 100.7496 degrees. The effective coefficient is 17.11066. These magnitudes escape the .309 lower bound of the orthogonal branch, but their ordering is not the observed Cabibbo hierarchy. They are a local capability example, not a data fit. The independent commutator determinant is nonzero, about  $1.38e-34$  in this dimensionful normalization. CP conjugation preserves potential energy and magnitudes and reverses  $J$ .

The nonzero source  $F$  backgrounds also require a matter stability check. Their weighted Frobenius norm gives an upper bound approximately  $7.55e-5$  on the source-induced quark-scalar bilinear operator norm. The specified positive universal matter breaking mass squared  $1e-4$  leaves a rigorous positive lower bound approximately  $2.45e-5$  after adding the nonnegative supersymmetric mass block. This check concerns the displayed matter action; arbitrary additional matter EFT operators are not included. Electroweak vacuum selection remains separate.

## Shared heavy directions and the ordering of the mixing angles

A second illustration keeps the dominant source ray common to both sectors and varies the two lighter down-sector rays. There are 946 unordered pairs to inspect; 902 pass the declared determinant cutoff .05, and 467 give nonzero  $J$  with  $V_{ub} < V_{cb} < V_{us}$  at the stated quark weights. These are numerical background diagnostics, not a complete classification of all nonorthogonal triples or a proof of energy selection.

The smallest- $V_{us}$  pair within this explicitly ordered sample uses down rays 18,21,10. This selection is phenomenological and is reported as such; no measured anchor, target phase or golden coefficient is fitted. Fully minimizing the same 263-coefficient potential from that starting point gives  $|V_{us}|=.425322$ ,  $|V_{cb}|=.0216236$ ,  $|V_{ub}|=.0112854$  and  $J=6.19010e-5$ . Its stationarity residual is below  $9e-12$  and its raw Hessian minimum is 106.15. The phase is 41.2339 degrees and  $C_{\text{eff}}=1.22692$ . The largest angle remains too large and the coefficient is not golden. The source sectors are nonorthogonal and the shared heavy direction only becomes approximate after relaxation.

This demonstrates how shared dominant directions and hierarchical weights can suppress mixing involving the heavy family while retaining a larger light-family angle and physical CP. It

does not establish that the present symmetry explains either the weights or the measured hierarchy. The same potential has other local branches and no cosmological occupation rule. The example supplies a concrete test case for a proposed source-label interaction.

## Why nonorthogonality changes the protection problem

$$H \Delta q = -\nabla O, \quad \text{rank} \frac{\partial(|V_{us}|, |V_{cb}|, |V_{ub}|, J)}{\partial c} = 4$$

There is an exact elementary theorem behind the numerical sensitivity. For a fixed full-rank column matrix  $C$ , let  $H(t) = \sum_i t_i c_i c_i^\dagger$ , with all  $t_i$  independently positive. If its unordered nondegenerate left eigenprojectors stay fixed on an open parameter set, then differentiating  $H$  with respect to each  $t_i$  shows that every rank-one matrix  $c_i c_i^\dagger$  is diagonal in that same eigenbasis. They therefore commute. Two nonzero rank-one projectors commute only when their vectors are parallel or orthogonal; full rank excludes parallel columns. The columns must be pairwise orthogonal. Conversely orthogonal columns share their projectors throughout every fixed mass-order region.

Thus nonorthogonal sources solve one problem and introduce another obligation: the quark-column coefficients affect the mixing frame. The scalar potential on the zero-matter/Higgs slice does not determine those coefficients. This theorem applies directly to the universal up mediator and to both sectors on a scalar aligned link. On the deformed link the fully matched numerical calculation establishes the response. The theorem concerns the full frame; it does not by itself rule out a specially protected single observable identity at another exceptional configuration.

At the actual factor-one vacuum, calculate the response to six same-sector overlap coefficients, all nine  $X_{ij}$  and all six logarithmic quark weights. Scalar variations solve  $H \Delta x = -\text{grad } O$  using the full 54-dimensional Hessian. The resulting Jacobian of  $(|V_{us}|, |V_{cb}|, |V_{ub}|, J)$  has rank four. Because  $J$  is small, the raw matrix is poorly scaled. After dividing each observable by its nonzero reference magnitude and normalizing the parameter columns, its singular values are approximately 3.75815, 2.06946, 1.60725, .10196. The fourth direction is well separated from numerical roundoff in these coordinates.

The implicit function theorem supplies nearby scalar vacua, and the inverse function theorem applied to a nonsingular four-column observable minor supplies a neighborhood of physically independent observable values. On this regular local branch with the specified couplings free, an exact golden amplitude relation or a fixed CP phase therefore cannot be an identity. This is a local numerical-rank statement supported by explicit positive masses, not a theorem excluding every possible branch, new representation or correlated UV completion.

The stronger calculation holds the six quark singular values fixed. Write the six log-mass response as  $M_{\text{scalar}} \Delta c + M_{\text{weights}} \Delta \log \lambda$ . The weight block is invertible, with singular values between .9845 and .9999. Set  $\Delta \log \lambda = -M_{\text{weights}}^{-1} M_{\text{scalar}} \Delta c$  and substitute it into the mixing response. The remaining fifteen scalar directions still generate all four physical mixing directions. With the same fractional-observable and column normalization, their singular values are 3.26373, 1.59728, 1.33898, .06274. Thus the freedom is not removed by calibrating all six quark masses. These reference masses are the illustrative masses of this vacuum, not experimental measurements.

## Fully re-minimized finite deformations

$$C_{\text{eff}} = \frac{|V_{ub}|(1 - |V_{ub}|^2)}{|V_{us}||V_{cb}|}, \quad C_{\text{eff}}^2 - 3C_{\text{eff}} + 1 = 0$$

The nominated coefficient is  $C_{\text{eff}} = |V_{ub}|(1 - |V_{ub}|^2)/(|V_{us}||V_{cb}|)$ . The golden condition requires  $C_{\text{eff}}^2 - 3C_{\text{eff}} + 1 = 0$  with  $0 < C_{\text{eff}} < 1$ . At the retained vacuum, the golden polynomial has nonzero derivatives along independent allowed directions. This is not an artifact of an off-vacuum source frame or an unnormalized quark matrix.

Perturb the  $X_{02}$  coefficient by plus and minus one percent of its absolute value, and solve the stationarity equations again.  $C_{\text{eff}}$  changes from 17.11066 to approximately 18.22351 and 15.99458. Perturbing the  $B_{02}$  same-sector overlap gives approximately 15.88999 and 18.32730. All four fresh vacua have positive Hessians and stationarity residuals below  $2e-10$ . Their receipt contains full observables and stability diagnostics. These are dimensionful EFT perturbations of the explicitly chosen coefficient values, not experimental error ranges or estimates of radiative corrections.

After re-solving each vacuum, independently refit all six positive quark-column weights to recover the baseline six masses. Their maximum log-mass residual is below  $2e-12$ . The  $X_{02}$  deformations still give  $C_{\text{eff}}$  approximately 18.22682 and 15.99157; the  $B_{02}$  deformations give 15.89963 and 18.31621. Since the tree-level zero-matter/Higgs scalar potential is independent of these quark weights, this compensation preserves scalar stationarity. This supplies finite examples of observable change at fixed masses, in addition to the local differential-rank test.

## The next interaction must constrain both sectors

An unconstrained joint-potential search can now be distinguished from a flavor prediction. The declared field content admits nonorthogonal stable vacua, weak CP and hierarchy mechanisms. Its complete low-degree operator space and independent quark weights leave all four physical mixing parameters adjustable on the tested branch. Finding a golden-looking point in that space would select couplings rather than explain a relation.

The next construction needs a justified non-Abelian action on source labels or a UV interaction whose unique invariant contractions correlate the quark-column weights and the nine mixed scalar coefficients. It must retain its allowed norm-square operators and kinetic corrections in the audit. A larger Abelian phase group cannot forbid the  $X_{ij}$ . A proposed M22-derived interaction must specify which of these actual contractions it relates, rather than supply an arbitrary matrix on a multiplicity space. New fields are useful only if their full operator space reduces the demonstrated physical freedom. No specific new representation is asserted here to achieve that reduction.

## Verification and established context

Run `OPENBLAS_NUM_THREADS=1 PYTHONPATH=python python python/develop_valentiner_joint.py` to reproduce the exact multidegree census, complete coefficient list, four mixed-potential cases, CP partner, shared-heavy-ray witness, 21-parameter response and four freshly minimized deformations. The field coordinates and every independent scalar coefficient are retained. The tests independently check all operator covariance under family generators, all six shaping factors and CP, every directional derivative, an independently contracted link F potential, the shared-column CP determinant identity, full stationary spectra, finite deformation replay and the eigenframe weight theorem.

Cross-coupling-driven reorientation and flavon alignment are established model-building problems. Relevant primary comparisons are M. Holthausen and M. A. Schmidt, Natural Vacuum Alignment from Group Theory: The Minimal Case, <https://arxiv.org/abs/1111.1730> ; and I. de Medeiros Varzielas, M.-S. Liu, A. Sengupta and J. Talbert, Residual Symmetries and Scalar Multiplet Vacuum Alignment in Non-Abelian Flavour Models, <https://arxiv.org/abs/2512.19789> . Their group models differ from this construction. The present results are the explicit operator census, nonorthogonal branch calculations, CP invariant and local freedom test for this specified EFT. Publication novelty and robustness beyond the declared truncation require independent review.